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Real-Time Vehicle Object Detection Method Based on Multi-Scale Feature Fusion

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ABSTRACT Existing object detection algorithms are affected by scenes with poor robustness, besides the existing public datasets are not applicable to urban road traffic scenes. In order to solve the problems of low accuracy in detecting panoramic video images, high false detection rate, this article designed a real-time traffic information detection method based on multi-scale feature fusion. For a start, the vehicle equipped with hp-f515 driving recorder collected video under the real road scene in Beijing. The total length of the driving route was 11 km. Extracted the recorded video, divided the video into frames which size was 1920 pixel × 1080 pixel, the classification was based on the type of vehicles in the general road landscape, and the format of datasets was Pascal VOC. Subsequently, an improved SSD (Single Shot Multi Box Detector) detector was designed, used single-data deformation data amplification methods to perform color gamut transformation and affine change on the original data to generate new data types; utilized the learning rate-adaptive adjustment algorithm to improve the efficiency of detector training. Eventually, the detector was used to detect traffic information in actual road scenes. The experimental results were compared with other traditional detectors. Extensive detection experiments showed that the detector had a processing speed of 55.6ms/frame with an accuracy rate of 98.53%, which could accurately identify multiple objections, small-distant objections, and overlapping objections in actual road scenes. It could provide the advanced driving assistance system with the perception information of the surrounding environment in time.

INDEX TERMS Deep learning, object detection, single shot multi box detector, data augmentation, learning rate.

I. INTRODUCTION

In today's extremely rapid economic development, the vehicle has become the main means of transportation in people's daily life. Nowadays, with the increasing number of vehicles, the problems related to traffic safety and traffic congestion are becoming increasingly prominent. According to statistics [1], the number of deaths by traffic accidents in the world is about 1.2 million every year, and there are millions of people injured or even dis-abled due to traffic safety accidents. In recent years, the economic loss caused by traffic safety has reached nearly 500 billion US dollars. The main factors causing traffic accidents include vehicle safety factors, road environment factors, and driver factors, among which driver factors account for the largest proportion. It is understood that the number of traffic accidents caused by drivers in the UK is as high as 95%, the proportion in the United States reaches 94%, which is 90% in China. Therefore, a key research direction in today's automotive safety field is how to reduce traffic accidents caused by drivers. As the number of cars increases, the vehicles speed up, the efficiency of road operation decreases, and the traffic congestion increase. To gradually solve the above problems and reduce the degree of harm to society. The artificial intelligent vehicle assisted driving technology provides a new way, researchers are committed to researching a real-time vehicle detection and management method. With the great breakthrough of machine vision [2], [3], deep learning [4]–[6], image processing, and other research fields, intelligent transportation systems came into being. The intelligent transportation system has been widely used in intelligent transportation, driverless and other fields. As an important part of an intelligent transportation system, how to correctly identify the attributes of vehicles is a concern for re-searchers. Vehicle object detection is a specific task of object detection, researchers focus on two aspects: hardware and computer vision. In terms of hardware,

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it focuses on how to collect effective traffic information and process, transmit information through a sensor system. However, the cost of sensor manufacturing is high and cannot be mass-produced on a large scale. From the perspective of computer vision, the use of algorithms for video data processing, as long as a good algorithm is designed to improve detection accuracy and efficiency. So that it can be used on a large scale. Therefore, researchers carry out in-depth research on computer vision.

At present, due to the fast speed, different colors, shapes of vehicles, and the changes of complex factors. Such as mutual occlusion and illumination (weather) in the actual driving process, it becomes more difficult to select vehicle samples and detect vehicle objects. Researchers need to constantly improve the algorithm to enhance detection accuracy.

In this article, a real-time vehicle object detection method based on multi-scale feature fusion was considered. We designed the methods to cope with the loss of semantic information caused by light or bad weather, small vehicle objects, missed detection of obscured objects, and other challenges under complex road conditions. Ultimately improved the detection accuracy of real-time vehicle objects in complex traffic scenarios. The main contributions of this article are as follows: Used single-data deformation data augmentation methods to perform color gamut transformation, affine transformation on the image to obtain more plentiful data. Used the learning rate-adaptive adjustment algorithm to set the enhanced SSD detector.

The rest of this article is organized as follows. The section II introduces the related detection model. The section III illustrates the experiment preparation and methods. Next, this article introduces object tracking and results from analysis in section IV. The conclusion is given in Section VI.

II. RELATED WORKS

A. DATASET DEVELOPMENT

Data is the core of deep learning methods [2]; A large number of data sets are produced for algorithm evaluation [3]–[7]. We list some common data sets, which are widely used in object detection.

1) MS-COCO DATASET

Coco dataset [8] is large and rich object detection, segmentation, and subtitle data set, which collects data through extensive use of Amazon Mechanical Turk. This dataset is objected at scene understanding, which is mainly intercepted from complex daily scenes. So far, the largest dataset with semantic segmentation provides 80 categories, more than 330000 images, of which 200000 are labeled, and the number of individuals in the whole dataset is more than 1.5 million.

2) PASCAL VOC DATASET

Pascal VOC provides a set of standardized and excellent data sets for image recognition and classification. From 2005 to 2012, an image recognition challenge is held every year. This dataset can be used for image classification, object detection, and image segmentation. Pascal voc2007 [9] data set contains 2501 images for the training set, 2510 images for the verification set, and 4952 images for the test set. In Pascal VOC 2012 [10], these numbers increased to 23080 pictures [2].

3) CIFAR DATASET

Cifar is an 80 million small image data set collected by Alex krizhevsky, Vinod Nair, and Geoffrey Hinton of the Canadian Institute of advanced technology. It contains cifar-10 and cifar-100 data sets. Cifar-10 consists of 60000 32×32 RGB color images, a total of 10 categories. The biggest feature of this data set is that it can migrate recognition to ubiquitous objects and apply it to multiple classifications.

4) CALTECH DATASET

Caltech is the image database of Caltech, including caltech101, and caltech256, which was collected by Fei Feili, Marco Andreetto, Marc'Aurelio Ranzato in September 2003. Caltech 101 contains 101 kinds of objects, each with about 40 to 800 images, and most of them have about 50 images. Caltech256 contains 256 categories of objects, about 30607 images.

B. DEEP LEARNING

In order to make machine vision more adaptive in recent years, deep learning technology has been widely used in tasks such as object detection and image classification, which have made great achievements [11], [12]. In particular, the convolutional neural network algorithm can improve the robustness of object tracking [6], which has been widely used in signal recognition [13], traffic volume prediction [14]–[17], face recognition [18]–[21], security assistance [18], [22], [23], driving technology [24]–[27], and autonomous driving technology [27]–[30]. Based on the convolution network, the object detection algorithm is divided into two categories [31]: two-stage object detection algorithm and single-stage object detection algorithm according to whether candidate frames are generated. The two-stage object detection algorithm first extracts the candidate box from the image and then carries out the two corrections to get the detection result. Though its accuracy is high, it takes a long time to meet the real-time requirements. The single-stage object detection algorithm can obtain the detection results quickly, but the detection accuracy is low.

R-CNN (Region-based Convolutional Network) [29] candidate frame is obtained by a selective search algorithm, then the candidate region is extracted by a deep convolutional neural network. The classification information of the fixed-length eigenvector and the corresponding coordinate position information is obtained. It effectively solves the scarcity problem of training set data, however, the training speed is slow, there will be more loss of image information;

SPP-Net (spatial pyramid pooling network) is 24-102 times faster than the R-CNN method [32], but who needs to go through a plurality of stages with a poor deep network detection accuracy of training; Fast R-CNN uses the image and region of interest which were input to the basic convolution network to be aggregated into the feature map [33], the softmax probability, and class border-box(correct the border of the candidate area and adjust its position and size) are returned through the full link layer. It solves some of the shortcomings of R-CNN and SPP-Net, Unfortunately, the feature extraction of region proposal not only requires selective search but also can only run this algorithm on the CPU. Faster R-CNN inputs convolution features into the RPN network to get feature information of the candidate box and then uses a classifier to judge whether it belongs to a specific class [34]. Finally, for the candidate box belonging to a certain feature, the position of the candidate box is further adjusted by the regressor. It can run on GPU, which greatly improves the detection speed, but it cannot meet the real-time requirements.

The image adjusted by YOLO (You Only Look Once) is input into a single convolution network [35], and the detection result is thresholded by the confidence of the model. Yolo algorithm regards detection as a regression problem, which has a simple process and fast detection speed. However, it has weak detection ability and poor robustness in the face of close objects and small objects. SSD can make a complete prediction for objects with different resolutions combined with its multiple feature maps [36], can realize the task of multi-scale object detection, but it still has insufficient detection accuracy in the face of small objects.

In order to enhance the efficiency of small object detection, we propose an enhanced SSD detection algorithm, which can be used in real-time traffic information detection

III. SSD DETECTION MODEL

A. DEEP ENSEMBLE MODEL

We used an SSD detection method based on the VGG-16 network [16]. In order to detect objects of different sizes, the SSD selects 6 feature maps for detection, as shown in Figure 1, which are conv4_3, conv7, conv8_2, conv9_2, conv10_2, conv11_2, the input image needs convolution and size transformation, the formula for the object detection process is as follows:

$$R_n = \varphi_n(R_{n-1}) = \varphi_n(\varphi_{n-1}(\cdots\varphi_1(I))) \tag{1}$$

$$T = D(d_n(R_n), d_{n-1}(R_{n-1}), L, d_{n-k}(R_{n-j})) \tag{2}$$

$$S_k = S_{min} + (k-1)\frac{S_{max} - S_{min}}{m-1} \tag{3}$$

where R_n represents the feature map of the nth layer,

φ_n is the non-linear operation of the nth layer feature map calculated from n-1th layer's.

$\varphi_1(I)$ is the operation of the feature map which obtained the first layer from the input image I.

$d_n(\cdot)$ is the corresponding scale test result range based on the nth layer feature map, $D(\cdot)$ is the final result T obtained by collecting all the intermediate detection result.

In formula (3), S_{max} and S_{min} represent the maximum and minimum values of the ratio of the default box to the input image, m represents the number of feature maps. The aspect ratio of each feature map default box is 1, 2, 3, $\frac{1}{2}$, $\frac{1}{3}$, the default box with a ratio of 1 corresponds to different sizes, which are S_k and $S'_k = \sqrt{S_k S_{k+1}}$ respectively.

Feature extraction is abstracted layer by layer in the SSD model, low-level features are mainly for detailed information, high-level features are mainly for abstract semantic in-formation. It can be seen from Figure 1 that the mapping area on the feature layer Cov9_2 has exceeded half of the output image size. The low-level feature layer used for small object detection is only Cov4_3 whose feature expression abilities and detailed information is insufficient. The feature extraction of the medium object is not accurate in the detection of small objects. As a result, the model cannot meet the requirements of accurate detection of multiple vehicle objects in real-time.

Given the above-mentioned shortcomings, after investigation, it is found that data diversity, learning rate directly affect the accuracy of the detector model. So as to satisfy the needs of real-time and accurate detection of the vehicle safety driving assistance system. We propose the following two improvement methods: use single-data deformation data augmentation methods to augment the original database. A learning rate-adaptive algorithm is proposed to realize the adaptive adjustment of the learning rate during model training.

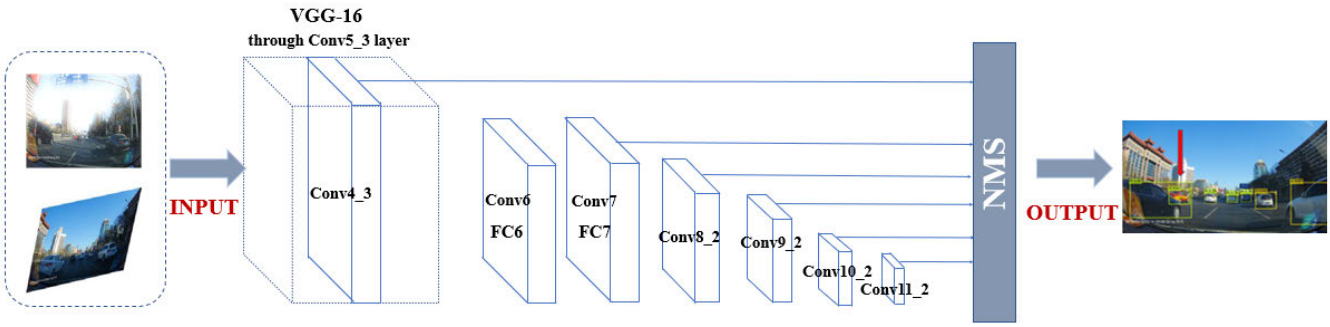

FIGURE 1. SSD network model.

B. DATA AMPLIFICATION

In the actual driving process, the intensity of the picture distortion is affected by the position of the light source, the reflected light of the glass, as well as the high beam illumination of the car at night. Since the overall change of the light source during the shooting process is not obvious, the acquired partial distortion samples account for less than the overall samples. So as to enrich the amount of training data, we propose the following two Single-data deformation data augmentation methods: through the color gamut trans-formation and the affine transformation operation of the space coordinates.

Through color gamut transformation, the matrix of image data is stacked, the bright-ness of each channel of the image is transformed in the light of the illumination to generate new samples. The elements in the matrix are the pixel values in the image. If the digital image of each frame of the captured video file is arranged in an array, the corresponding matrix of each channel is expressed as:

$$f(x,y)=\begin{bmatrix} f(0,0) & f(0,1) & L & f(0,N-1) \\ f(1,0) & f(1,1) & L & f(1,N-1) \\ M & M & ,\ldots, & M \\ f(M-1,0) & f(M-2,0) & L & f(M-1,N-1) \end{bmatrix} \tag{4}$$

$f(x,y)$ represents the pixel value of each pixel. Establish a linear map to adjust the image in the original data set, the formula is expressed as:

$$Y(x,y)=aX(x,y)+b \tag{5}$$

In formula (5), $Y(x,y)$ represents the object pixel value; $X(x,y)$ is the source pixel value; α is the slope; β is the intercept.

$\alpha > 1$, the contrast of the image can be enhanced, not only the pixel values of the image all become larger after the transformation, but also the overall brightness is increased, as shown in Figure 2.

$\alpha = 1$, it is often used to adjust image brightness.

$0 < \alpha < 1$ is opposite to the situation of $\alpha > 1$, the image pixel values all become smaller after the transformation, accordingly the whole is weakened.

The driving recorder is located in the center of the vehicle to simulate different driving conditions. For example, when the vehicle is turning left or right, the angle of the passing vehicles in the driver's field of view will change. Therefore, an affine transformation is used to reposition the pixel to a new position. The expression is as follows:

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \boldsymbol{T}\begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} a & b & c \\ d & e & f \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \tag{6}$$

Affine matrix

$$T=\begin{bmatrix} 1 & 0 & 0 \\ \tan\varphi & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

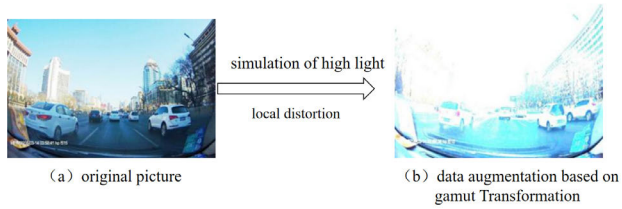

FIGURE 2. Schematic diagram of data augmentation.

$(x', y', 1)$ Represents the pixel coordinates in the original image, $(x, y, 1)$ represents the transformed pixel coordinates, φ is the transformed angle, as shown in Figure 3.

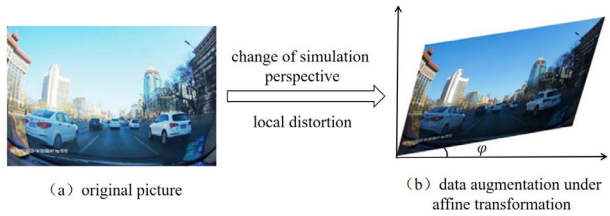

FIGURE 3. Schematic diagram of data augmentation.

Figure 2(a) is the original image data. Figure 2(b) is the data augmentation for the influence of light factor, which is to change the brightness of the original image to simulate the data collection under the condition of relatively large changes in illumination and obvious brightness. Figure 3(b) is for the image data distortion, simulating the image deformation caused by the change of the driver's field of view or other external influence factors when the left or right steering occurs. Data augmentation can effectively enrich the amount of training data and help train a more generalized network.

C. ADAPTIVE ADJUSTMENT METHOD OF LEARNING RATE

A higher learning rate will result in significant fluctuations in the loss value, while overfitting will lead to unstable model performance; a lower learning rate is not conducive to the rapid development of the detector model. To prevent the above situation, we proposed an optimal algorithm for adaptive adjustment of the learning rate in the training process. The core idea of the learning rate-adaptive adjustment algorithm was to use the change of training times to judge the loss value and set the learning rate until the model performance becomes stable, as shown in Table 1.

When using the learning rate adaptive method, we can roughly divide it into four categories: 0-20000 times, 20000-40000 times, 40000-80000 times, 80000 times and above. Through the experimental results, it is found that the loss value gradually decreases with the increase of training times. When the training times reach 20000 times, the loss value is less than 1.0; When the training times reach 60000 times, the loss value tends to be stable, and the value is 0.5, that is, the performance of the model tends to be stable. According to the change trend of the loss curve of

TABLE 1. Adaptive adjustment algorithm of learning rate.

Algorithm1 Adaptive adjustment algorithm of learning rate

Input: Learning rate LR = 0.001, training num = 0;
Output: Loss value, training num

```
while Train num ≤ 20000;
    if loss ≥ 2.5
        lr=0.0001
while 20000 < Train num ≤ 16000
    if loss ≥ 1.6
        lr=10E-5
    else
        lr=0.0001
while 40000 < Train num < 80000
    if loss ≥ 0.8
        lr=10E-6
    else
        lr=10E-5
while Train num > 80000
    if loss ≥ 0.5
        lr=5 × 10E-6
    else
        lr=10E-6
```

the improved SSD detector model trained at present, the performance of the model has been stable after 60000 times of training. In order to shorten the development time of the model, 40000 times, 60000 times, 80000 times, 100000 times and 120000 times of training were selected for testing, and the recall value, precision value and false rate value of the five detector models were calculated respectively, as show in Table 2.

TABLE 2. Five training times model detection and evaluation indexes.

Training num (Ten thousand times)	4	6	8	10	12
Recall (%)	62.97	73.68	91.53	97.67	98.53
Precision (%)	76.28	79.63	87.40	93.91	98.80
False rate (%)	20.04	18.72	16.36	16.23	13.70

To sum up, recall value and precision value gradually increase with the increase of training times, and false rate value gradually decreases, with the increase of training times, recall rate and accuracy gradually increase, and false detection rate gradually decreases. It can be seen that the higher the training times, the better the detector effect. At the same time, in order to improve the development efficiency of detector model, 120000 times of model training has reached the detection effect required by this subject. Therefore, 120000 times of improved SSD detector model training is preliminarily determined.

IV. EXPERIMENTS AND RESULTS BAIDU FANYI

A. PRODUCTION OF DATASETS

The HP-F515 1080P driving recorder was mounted on the vehicle as an experimental platform. Real vehicle experiments were conducted on real urban road scenes, including urban arterial roads, bridge highways, signal junctions, etc. The driving distance was 11km. As shown in Figure 4, the actual vehicle experiment was arranged at 3:00-5:00 in the afternoon and 7:00-9:00 in the evening. During this period, the test route had a large traffic volume with some congested sections.

Recorded 123 videos totaling 400 minutes, framed the video and set the size to 1920pixel × 1080pixel. According to the vehicle object types in the normal highway scene, the vehicle objects were divided into three types, as shown in Table 3. They were large vehicles (including buses, trucks, etc.), medium vehicles (including vans, off-road vehicles, flat-tail vehicles, etc.) and small cars (including cars, etc.)

TABLE 3. Three objects and their labels.

NO.	Object vehicle type	Description
1	compact car	cars etc.
2	medium-sized car	Vans etc.
3	large vehicle	Buses etc.

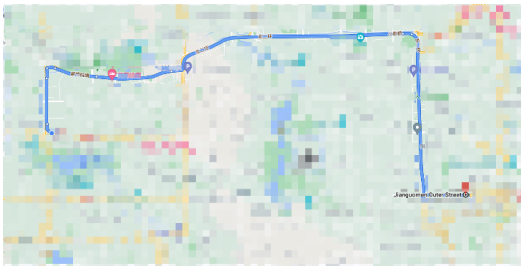

FIGURE 4. Road map of real vehicle experiment.

Through the Pascal VOC datasets format, we used Lambimg to make a training data set and a test data set. Selected 92 videos and added them to the training set in the ratio of day: night = 5:1. Selected another 81.1MB video as the test data set. Framed the video, the training set contains 12614 photos, the test set contains 5400 photos. There was no overlap between the test data set and the training data set.

B. QUANTITATIVE PERFORMANCE EVALUATION

1) COMMON VOLUME NEURAL NETWORK TRAINING AND DETECTION TIME COMPARISON

Compared with Fast RCNN, traditional SSD had a lower accuracy rate, compared with other methods, traditional SSD detection algorithms cannot better identify small objects. In order to further verify the practicability of the enhanced SSD model, three commonly used object detection models and the enhanced SSD detection model were selected for performance comparison analysis. The detection models used for comparison were Fast RCNN detector based on VGG-16, Faster RCNN detector based on ResNet50, YOLOv3 detector based on DarkNet53, and YOLOv5 detector. The experimental verification process used four types of detectors to detect the image information of the same content and calculated the test time of the three detectors. The test results were shown in Table 4.

TABLE 4. Comparison of training and detection time of several algorithms.

Algorithmic model	Fast RCNN	Faster RCNN	YOLO	enhanced SSD
Training time (h)	82.5	40.5	38	47.3
Number of training sessions (tens of thousands)	12	12	12	12
Video test frame rate (FPS)	3	5	7	18
Picture test time (ms)	142.9	128	73	55.6

The Table 4 shows that the Fast RCNN algorithm takes the longest time in video detection, and image testing. The Faster RCNN algorithm belongs to an end-to-end network structure, so compared to the Fast RCNN algorithm, it reduces the GPU occupancy and calculation, but its detection time is longer. The two detection algorithms, YOLO series detection algorithms and the enhanced SSD are superior to the other two detection algorithms in terms of video and image detection speed, which can meet real-time requirements. Therefore, in subsequent experimental tests, only the enhanced SSD algorithm and the YOLO algorithms are compared and analyzed.

2) COMPARISON OF ENHANCED SSD AND YOLO DETECTION RESULTS

The detection video contained 5400 frames of images, which were divided into three types of objects: large vehicles, medium vehicles, small vehicles. The detectors including YOLOv3, YOLOv5 enhanced SSD, used them to detect and recognize the video (The models are all trained by Titan X graphics card 120,000 times) the relevant calculation results are shown in Figure5.)

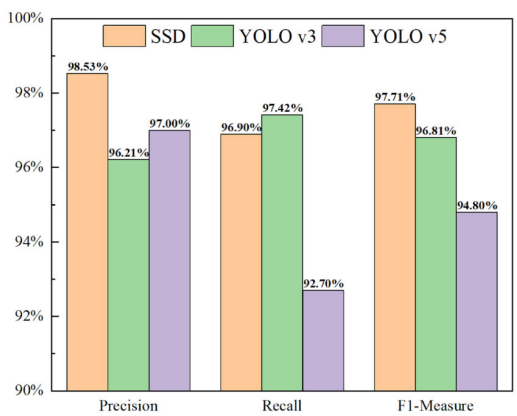

FIGURE 5. Comparison of the enhanced SSD and Yolo detection results (a, precision, recall, F1 measure detection effect; b. Flame, misrate detection effect).

Considering the experimental results of five performance indicators, the enhanced SSD detector is more accurate and better than the YOLO series detector in the performance of Precision, F1-Measure.

C. QUALITATIVE PERFORMANCE EVALUATION

In this paper, the video included complex scenes such as multi-object, remote small objects, and overlapping objects in a real road environment. In order to simulate the detection effect of the YOLO series detector and enhanced SSD detector on real road conditions, the main challenges of video object detection were listed in table form, and the detection effect was compared.

1) COMPARISION OF ENHANCED SSD DETECTOR AND YOLOV3 DETECTOR DETECTION RESLUTS

The main challenges in video object detection are shown in Table 5. In order to facilitate comparison, we annotated the detection objections that the detector needed to compare with arrows.

TABLE 5. Main video challenges.

Number of Frames	Main challenges
152	Large object blocked in the distance
1857	Near incomplete object
2790	Small distant objects in dim light
3184	Occluded small object in the distance
3941	multiple object
3985	multiple object

As shown in the 152nd frame in Figure 6 and Figure 7, there is an occlusion of a small object vehicle by a nearby large object. The enhanced SSD detector detects seven vehicle objects: two medium-sized cars, five small cars, successfully detects the vehicle object pointed by the arrow and the positioning is accurate.

When the YOLOv3 detector is blocked during the tracking process, the tracking effect changed drastically. The 1857th frame in the video shows that during the tracking process, the enhanced SSD detector can detect all five objects: four medium-sized cars, and one small car. The detection accuracy is higher, it can detect incomplete objects nearby with a more beneficial functioning on-board safety-assisted driving system; the YOLOv3 detector only detects four objects: two medium-sized cars, and two small cars, but the object pointed by the arrow is not detected. Therefore, the YOLOv3 detector cannot detect the object category accurately, especially similar objects such as medium-sized cars and small cars.

As shown in the 2790th frame, the result of the enhancing SSD detector is five objects detected, include three medium-sized cars, two small cars, and can detect the small object pointed by the arrow in the distance, when the distance small object appeared. However, compared with the enhanced SSD detector, the YOLOv3 detector only can detect the four objects: two medium-sized cars, two small cars, the distance small object is also undetected. In the 3184th frame, the enhanced SSD detector detect four objects include three small cars and a large car, besides, the small car which far away, blocked, and pointed by the arrow is detected. On the other hand, only three objects are detected by the YOLOv3 detector, which includes two small cars and a large car, the distance small object is also undetected.

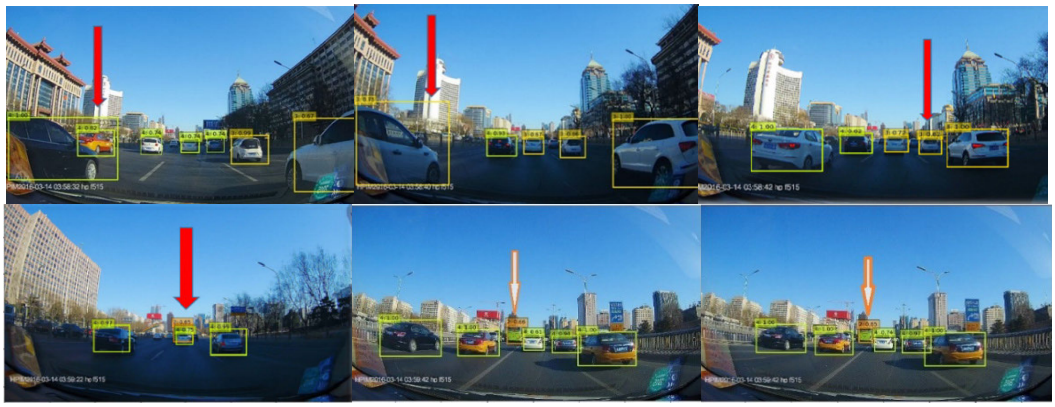

FIGURE 6. Enhanced SSD algorithm tracking effect.

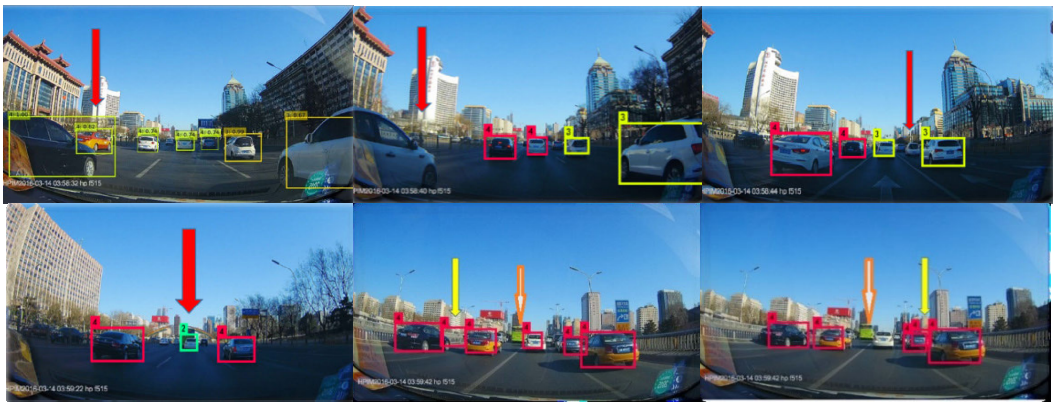

FIGURE 7. Tracking effect of Yolo algorithm.

According to the 3941st frame and 3985th frame, the YOLOv3 detector identifies the non-object object which pointed by the arrow as the small car, identifies the arrow point small car as a double small car, the far-away bus pointed by the arrow is also undetected. On the contrary, the enhanced SSD detector can detect the far away from a small car and the bus pointed by the arrow. The phenomenon of false detection also do not appear. Thus, the enhanced SSD detector cannot cause the phenomenon of false detection and missed detection, but the YOLO detector is the opposite.

In conclusion, the enhanced SSD detector and YOLOv3 detector track the same object in the video, and the enhanced SSD detector has a better detection effect for multi-objective, distant small objects and overlapping objects than the YOLOv3 detector, thus further verifying the correctness and accuracy of the enhanced SSD detection algorithm.

2) COMPARISION OF ENHANCED SSD DETECTOR AND YOLOV5 DETECTOR DETECTION RESLUTS

The main challenges in video object detection were shown in Table 6. In order to facilitate comparison, we annotated the detection objections that the detector needed to compare with arrows.

The 413st frame, the 3273rd frame, and the 3573rd frame in Figure 8 and figure 9 show that there are multiple near objects repeated detection in the video frame, it is extremely unfavorable to the vehicle safety assistant driving system with early warning function. As shown in frame 1651st, when a distant occluded object appears, the enhanced SSD detector detects four objects, including three small cars, and one large car; while the YOLOv5 detector can only detect three objects accurately, except the small objects sheltered.

TABLE 6. Main video challenges.

Number of Frames	Main challenges
413	Vicinity complete object
3273	Vicinity complete object
3575	Vicinity complete object
1651	Occluded small object in the distance
1620	multiple object
3124	multiple object

In the 1620th frame, the detection result of the SSD detector is four objects: three small vehicles, and one large vehicle. There is a leak detection for the small objects in the distance, while the YOLOv5 detector can detect four small vehicles, but cannot detect the small blocked objects. According to 3124th frame image, the SSD detector can detect three medium-sized vehicles and two small vehicle objects under complex road conditions, while the YOLOv5 detector detects two medium-sized vehicles and two small vehicles, which has some missing rate for the complete objects under complex road conditions.

To sum up, the enhanced SSD detection will not have serious false detection and missed detection, while the

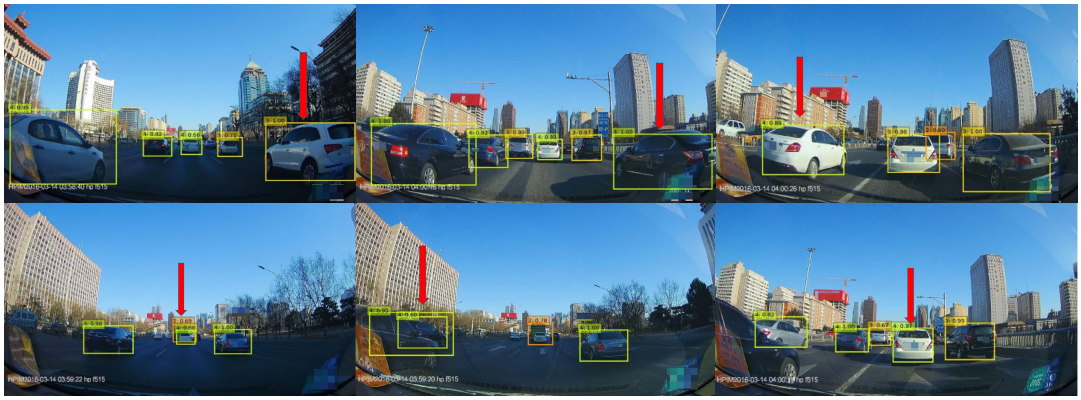

FIGURE 8. Enhanced SSD algorithm tracking effect.

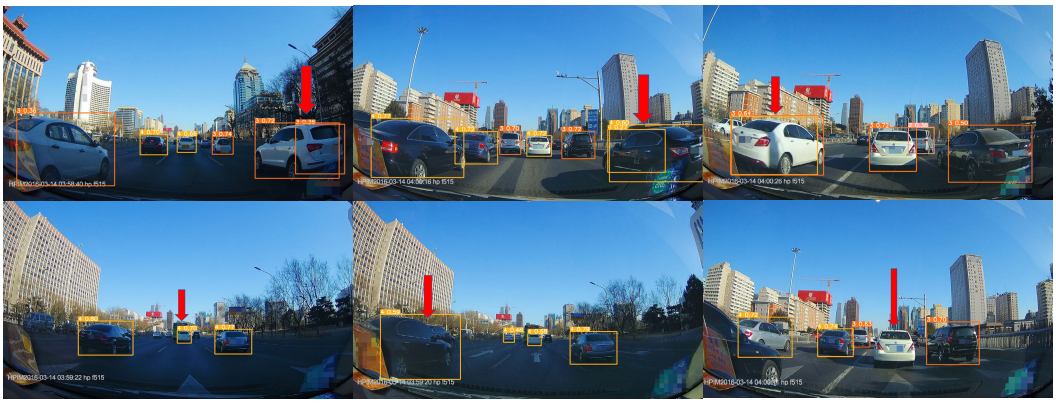

FIGURE 9. Tracking effect of Yolo algorithm.

YOLOv5 detector will still have obvious false detection and missed detection.

V. CONCLUSION

In this paper, the real-time vehicle object detection problem was studied. In order to simulate the real traffic situation in the city, the real shot video of the driving recorder was used as the data set of the detection algorithm. Enriched the diversity of data through data augmentation algorithms. The learning rate adaptive adjustment algorithm was used to replace the traditional and fixed learning rate, and it is applied to the model training process to generate a vehicle multi-object detector.

Finally, through experiments, the performance of the enhanced SSD detection algorithm was compared with the traditional detection algorithm. The experimental results showed that the detection time of the enhanced SSD detector is 55.6 MS, with a better detection speed than other traditional detection algorithms under the same resolution. Especially for multi-object, remote small object, and overlapping objects, the detection effect is good, which makes up for the shortcomings of the traditional detection methods, such as false detection and missing detection, and meets the real-time detection needs of practical application.

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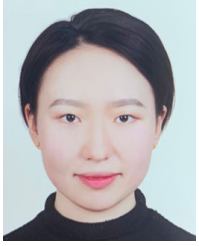

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